



Rhus chinensis Mill. fruits prevent necrotizing enterocolitis in rat pups via regulating the expressions of key proteins involved in multiple signaling pathways

Yi Zhang^a, Ou Wang^b, Hongying Mi^c, Junjie Yi^a, Shengbao Cai^{a,*}

^a Faculty of Food Science and Engineering, Kunming University of Science and Technology, Kunming, 650500, Yunnan Province, People's Republic of China

^b National Institute for Nutrition and Health, Chinese Center for Disease Control and Prevention, Beijing, 100050, People's Republic of China

^c The First People's Hospital of Yunnan Province, Affiliated Hospital of Kunming University of Science and Technology, Kunming, 650032, Yunnan Province, People's Republic of China

ARTICLE INFO

Keywords:

Apoptosis
Bowel disease
Chinese sumac
Dietary therapy
Intestinal histopathology
Inflammation

ABSTRACT

Ethnopharmacological relevance: Based on ancient records and previous studies, many parts of *Rhus chinensis* Mill., including the fruits, have good preventive and therapeutic effects on inflammation, malaria, diarrhea, and gastrointestinal diseases. *Rhus* plants and *Galla chinensis* produced from *R. chinensis* leaves can also prevent or cure intestinal diseases. However, the preventive effect and molecular mechanisms of *R. chinensis* fruits on necrotizing enterocolitis (NEC) have not been comprehensively studied.

Aim of the study: This article aims to estimate the effect of the 80% ethanol extract of *R. chinensis* fruits (RM) on alleviating NEC in rat pups and illustrate the potential molecular mechanisms.

Materials and methods: Rat pups were subjected to formula feeding, intermittent hypoxic, and cold stresses to establish the NEC model. The preventive effects of RM on NEC were evaluated through survival rate; clinical sickness index; macroscopic conditions; histopathology; and expression levels of inflammatory markers (i.e., tumor necrosis factor- α [TNF- α], interleukin-6 [IL-6]), oxidative stress indicators (i.e., total antioxidant status [TAS], total oxidant status [TOS], superoxide dismutase [SOD], glutathione peroxidase [GSH-Px], myeloperoxidase [MPO], malondialdehyde [MDA]), and tight junction proteins (i.e., Zonula Occludens 1 [ZO-1], Occludin). Moreover, the expression levels of several key proteins involved in oxidative stress (i.e., nuclear factor erythroid 2-related factor 2 [Nrf2], NAD(P)H-quinone oxidoreductase-1 [NQO1]), inflammation (i.e., Toll-like receptor 4 [TLR4], phosphorylated-nuclear factor kappa-B [p-NF- κ B]), inducible nitric oxide synthase [iNOS]), and apoptosis (i.e., cleaved cysteinyl aspartate specific proteinase-3 [cleaved Caspase-3], Bcl-2-associated X [Bax], B-cell lymphoma-2 [Bcl-2]) in intestinal tissues were analyzed to clarify the molecular mechanisms.

Results: The extract particularly high doses (400 mg RM/kg body weight) could remarkably reduce the mortality and clinical sickness score and improve the macroscopic condition and histopathological injury of the intestine in NEC pups. After RM administration, the levels of TOS, TNF- α , IL-6, MPO, and MDA in the bowel tissue decreased, whereas the levels of TAS, SOD, and GSH-Px were significantly enhanced. The expression levels of ZO-1 and Occludin proteins were dramatically augmented in RM-treated groups to maintain intestinal barrier integrity. Further analyses revealed that RM might prevent NEC pups by improving some pivotal proteins involved in oxidative stress, inflammation, and apoptosis of enterocytes, namely, by down-regulating the levels of TLR4, p-NF- κ B, iNOS, cleaved Caspase-3, and Bax and up-regulating the levels of Bcl-2, NQO1, and Nrf2.

Conclusions: The RM prevented the intestinal inflammation and damage caused by NEC by regulating the expression of several pivotal proteins involved in oxidative stress, inflammation, and apoptosis. This study might provide a scientific basis for *R. chinensis* fruits as a traditional herbal medicine to prevent and/or alleviate NEC.

* Corresponding author.

E-mail addresses: caikmust2013@kmust.edu.cn, caikmust2013@163.com (S. Cai).

<https://doi.org/10.1016/j.jep.2022.115103>

Received 24 December 2021; Received in revised form 3 February 2022; Accepted 9 February 2022

Available online 11 February 2022

0378-8741/© 2022 Elsevier B.V. All rights reserved.

1. Introduction

Necrotizing enterocolitis (NEC) is a serious gastrointestinal disease commonly seen in newborns, and its incidence and mortality in premature infants are as high as 12%–20% and 40%, respectively (Drucker et al., 2018). NEC is manifested as appetite loss, weight loss, abdominal distension, bloody diarrhea, and intestinal mucosal necrosis, causing severe systemic sepsis and multiple organ failure (Yang et al., 2020). Immature intestinal development, impaired intestinal barrier, formula feeding, inflammation, hypoxia, and colonization of pathogenic bacteria are considered as the main elements in the pathogenesis of NEC (Cigsar et al., 2020; Isik et al., 2019). Although NEC is the most commonly acquired gastrointestinal disease in newborns, methods to prevent or treat NEC, including antibiotic treatment, nasogastric decompression, and total parenteral nutrition, are currently limited (Neu, 2020). However, these methods have many side effects. Long-term parenteral nutrition can lead to intestinal atrophy, increased permeability, increased inflammation, and late-onset sepsis (Moss et al., 2008). Antibiotic use can cause dysbiosis in the gastrointestinal tract. Dysbiosis affects the development of the immune system, thereby leading to intestinal inflammation, and increases drug resistance (Van Belkum et al., 2020). Surgery should be considered if the patient does not respond to these medical interventions or develops bowel perforation. At present, more than 50% of cases require surgical removal of the necrotic intestine, which may severely affect the growing development and physical health of newborns (Niño et al., 2016). Therefore, a safe and effective treatment from traditional medicine should be exploited to prevent the onset of NEC.

At present, some natural active components from traditionally used medicinal plants have been proven to prevent NEC. For example, ginger (*Zingiber officinale* Roscoe) and fennel (*Foeniculum vulgare*) seed extracts can treat and prevent NEC by reducing mortality, clinical sickness score, histopathological injury, oxidative stress, inflammation, and apoptosis response in pups (Cakir et al., 2018; Yakut et al., 2020). *Rhus chinensis* Mill., a member of the genus *Rhus*, is also called Chinese sumac and widely distributed in Asia, including China, Japan, and India (Djakpo and Yao, 2010). Based on ancient records and previous studies, many parts of *R. chinensis*, including fruits, have good preventive and therapeutic effects on inflammation, malaria, diarrhea, and gastrointestinal diseases (Djakpo and Yao, 2010; Heirangkhongjam and Ngaseppam, 2018). The *Galla chinensis* generated by the *Schlechtendalia chinensis* on *R. chinensis* leaves can effectively treat malaria and diarrhea (Chen et al., 2006; Djakpo and Yao, 2010). Previous literature also proved that *Rhus* plants can effectively prevent or cure intestinal diseases. For example, *R. javanica* and *R. semialata* fruits prominently relieved the occurrence and severity of diarrhea (Bose et al., 2008; Tangpu and Yadav, 2004). *R. coriaria* has an evident preventive effect on NEC (Isik et al., 2019). Our previous study verified that 80% ethanol extract of *R. chinensis* fruits (RM) exhibits a good preventive effect on ulcerative colitis induced by dextran sulphate sodium (DSS) (Zhang et al., 2021). Therefore, *R. chinensis* fruits may also effectively prevent and/or control NEC, and this speculation needs to be verified by a scientific and comprehensive experiment.

According to previous reports, the combined treatment of formula feeding, intermittent hypoxic and cold stresses is a common method to establish the NEC model in rat pups (Lu et al., 2014; Sodhi et al., 2008). In addition, the rat NEC model is widely used due to its low price, large litter size, and easy availability (Lu et al., 2014). Therefore, in this work, rat pups are used as experimental subjects, and the NEC model is established in this study by formula feeding, hypoxia, and cold stress to investigate the preventive effect and underlying mechanisms of RM on NEC. This study may provide a scientific basis for further using *R. chinensis* fruits to develop a treatment method with high safety and few side effects for NEC prevention or alleviation.

2. Materials and methods

2.1. Materials and reagents

The BCA kit was obtained from Dalian Meilun Biotechnology Co., Ltd. (Dalian, China). The total antioxidant status (TAS), total oxidant status (TOS), tumor necrosis factor (TNF)- α , and interleukin (IL)-6 ELISA kits were obtained from Shanghai Enzyme-Linked Biotechnology Co., Ltd. (Shanghai, China). Malondialdehyde (MDA), superoxide dismutase (SOD), glutathione peroxidase (GSH-Px), and Western and IP cell lysis buffer including multiple inhibitors (phosphatase and proteases inhibitors) were obtained from Beyotime Biotechnology Co., Ltd. (Shanghai, China). Myeloperoxidase (MPO) was provided by Nanjing Jiancheng Biological Engineering Research Institute Co., Ltd. (Jiangsu, China). All the antibodies used in the experiment were obtained from ABClonal Technology Co., Ltd. (Wuhan, China). Phenolic standards (purity $\geq 97\%$) were obtained from the Joint Laboratory of Standard Sample Development of Traditional Chinese Medicine (Chengdu, China).

2.2. Preparation and analysis of samples

R. chinensis fruits were purchased from Kunming Plant Classification Biotechnology Co., Ltd. (obtained from Tengchong County, Baoshan District, Yunnan Province) in September 2020 and placed at -20°C . The fruits were identified by Dr. Y.P. Liu of the Kunming Institute of Botany, CAS, and a specimen of the fruits (No. kust20200925-03) was kept in the School of Food Science and Engineering, Kunming University of Science and Technology. The plant name was confirmed online (<http://www.theplantlist.org/tpl1.1/record/kew-2422836>). The freeze-dried fruits were ground with a high-speed crusher (Lingdan LD-T300, Shanghai, China), and the material filtered through an 80-mesh sieve was extracted twice with 80% ethanol to gain the extract (Ma et al., 2022; Wu et al., 2019). The main components of RM were separated by using a Thermo Fisher Ultimate 3000 UHPLC system (Thermo Fisher Scientific, Bremen, Germany) coupled with the Agilent Zorbax SB-C18 column (1.7 μm , 2.1 mm $\times$ 100 mm). Acidified water (0.1% formic acid) and acetonitrile were used as mobile phases A and B, respectively. Gradient elution conditions were as follows: 0–2 min, 5% B and 2–12 min, 5%–20% B. Solvent gradients were set on the basis of our previous report (Zhang et al., 2021) with minor modification. RM components were distinguished by using high-resolution mass spectrometry (Q-Exactive Focus) in negative mode with an ESI ion source (Thermo Fisher Scientific, Bremen, Germany). Mass spectrometry parameters were set identically as in our previous study (Zhang et al., 2018).

2.3. Animal experiment design

Five Sprague–Dawley rats that were 15 days pregnant were purchased from the Animal Experiment Center of Kunming Medical University (Kunming, China, SCXK Dian K2020-0004). Rats were placed in different cages, fed with maintenance feedstuff, and raised in a standard condition ($23^{\circ}\text{C} \pm 2^{\circ}\text{C}$, 40%–75% humidity, 12 h/12 h light/dark cycle). The rats gave birth to 49 pups (9, 9, 10, 10, and 11 for rats 1–5, respectively) spontaneously. Newborn pups were kept with their mothers and freely breastfed. The rat pups were grouped in accordance with the method of Gong et al. (2020). On the third day of birth, eight pups from each litter were randomly divided in accordance with weight into four groups ($n = 10$ for each group): control (breastfeeding), NEC (induced by NEC procedure), NEC + L-RM (induced by NEC procedure and treated with 200 mg RM/kg body weight), and NEC + H-RM (induced by NEC procedure and treated with 400 mg RM/kg body weight) groups. The doses of extract and the NEC procedure were determined on the basis of preliminary experiments and previous studies (Blackwood et al., 2017; He-Yang et al., 2020; Zhang et al., 2021). The whole experiment was carried out in strict accordance with the

Guidelines for Care and Use of Laboratory Animals of Kunming University of Science and Technology, and experiments were approved by the Animal Ethics Committee of Kunming University of Science and Technology.

2.4. NEC procedure and sample treatment

The pups in the control group together with their mother were freely breastfed and were not subjected to any procedure. The pups in the NEC and NEC + RM groups were separated from the mother rat and kept in an incubator with the same environmental conditions as the control group. Those three groups of pups were raised using a specific rodent formula (15 g Similac 60/40 [Abbott Nutritional, Columbus, OH] mixed with 75 mL Esbilac canine milk replacer [PetAg Inc., Hampshire, IL]) (Blackwood et al., 2017) in 0.2 mL every 4 h. If tolerated, the volume of feed formula was augmented by 0.1 mL per day (Blackwood et al., 2017; Isik et al., 2019). At the same time, the pups in the NEC, NEC + L-RM, and NEC + H-RM groups were all subjected to hypoxia and cold stress. Pups were exposed to hypoxia (95% N₂ and 5% O₂) for 10 min and immediately placed in a low-temperature environment (4 °C) for 5 min. This operation was performed twice a day for three consecutive days to induce NEC (Blackwood et al., 2017; He-Yang et al., 2020).

In this work, the pups in the NEC + L-RM and NEC + H-RM groups were orally administered with RM once a day through a feeding catheter (1.9 French PICC). The doses of RM in the NEC + L-RM and NEC + H-RM groups were 200 and 400 mg/kg (dissolved in normal saline), respectively, and the gavage volume was 2 mL/kg/day. The pups in the control and NEC groups were given normal saline (2 mL/kg/day).

2.5. Assessment of clinical sickness score and follow-up

The pups' behaviors were estimated by participants without knowing the grouping by using a rat pup's clinical sickness score. Scores ranged from 0 to 12, and parameters included appearance, touch response, natural activity, and body color (Table 1) (Zani et al., 2008). In the NEC procedure, the number and causes of death, including iatrogenic deaths, in each group were recorded every day.

2.6. Macroscopic and histopathological evaluation

On the fourth day of the study, all rat pups were sacrificed. The abdomens of pups were dissected, and the condition of the intestines was observed. The consistency, color, and dilation of the intestines were macroscopically evaluated (Table 2) (Zani et al., 2008). Subsequently, the intestinal tube between the lower end of the duodenum and the

Table 1
Clinical sickness score for pups.

Score	Appearance	Response to touch	Natural activity	Body color
0	Tonic and well hydrated, glossy skin	Alert without any stimulation	Moving normally in the cage	Pink
1	Slimmer, but still tonic and hydrated	Responding to mild stimulation (gentle foot touch)	Able to wriggle if put in the supine position	Pale or dark extremities, fingers/toes
2	Skinny, floppy, and dehydrated	Responding to vigorous stimulation (foot pinch)	Unable to wriggle if put in the supine position	Paleness on the whole body
3	Gasping and in agony	Unresponsive even to vigorous stimulation (does not become alert when picked up and moved)	Not moving limbs and lying still, supine or lateral decubitus	Gray

Table 2

Macroscopic assessment score of intestines.

Score	Consistency	Color	Dilation
0	Normal, does not tear when dissecting from surrounding tissue	Normal, yellow/tan	Normal, no dilation
1	Moderately friable	Patchy discoloration, pink, congested	Patchy areas of dilation
2	Liquefied, jelly-like, extremely friable	Extensive discoloration, dark, necrotic	Extensive dilation throughout small bowel

ileocecal area was excised, and about 1 cm of the distal ileum was collected for histopathological analysis. The remaining tissues were washed with normal saline to remove the contents of the intestines, prefrozen with liquid nitrogen, and stored at −80 °C. The distal ileum tissue was fixed with paraformaldehyde (4%), embedded in paraffin, and sliced into 4 μm sections, which were deparaffinized and subjected to hematoxylin–eosin (H&E) staining. Sections were observed at 200 × and 400 × magnifications under the Olympus IX83 microscope (Tokyo, Japan). The histological score system published by Zani et al. (scores = 0 to 4: 0, normal intestine; 1, disordered arrangement of intestinal villus cells and separation of villus–core; 2, significantly disordered arrangement of villus intestinal cells, villus–core separation down sides of villi, and blunting of villi; 3, epithelial sloughing of villi and loss of villi; and 4, necrosis) was used to assess NEC (Zani et al., 2008). A score of 2 or higher indicated the presence of NEC, and a score of 3 or higher indicated severe NEC.

2.7. Biochemical indicators

The levels of TAS, TOS, MPO, MDA, SOD, GSH-Px, TNF-α, and IL-6 were detected using the corresponding kit in accordance with the manufacturer's instructions. The intestinal tissue homogenate used to analyze the above indicators was prepared as follows. The intestinal sample was weighed, homogenized with NaCl solution (0.9%) at 1:9 (w/v), and the homogenate sample was centrifuged at 10 000 g and 4 °C for 7 min. Finally, the collected supernatant was stored at −80 °C for subsequent assays.

2.8. TUNEL analysis

The slices of the intestinal wax block were successively deparaffinized, rehydrated with graded alcohol, and incubated with TUNEL reagent at 37 °C for 2 h without light. Then, 4,6-diamino-2-phenylindole was loaded into the nuclei of intestinal cells. Finally, TUNEL-positive signals were observed using the Olympus IX83 microscope (Wu et al., 2020).

2.9. Immunohistochemical and immunofluorescence analyses

TLR4 and p-NF-κB expression levels in the intestine were measured by immunohistochemical analysis, and ZO-1, Occludin, and nucleus-Nrf2 expression levels in the intestine were measured by immunofluorescence analysis. Immunofluorescence and immunohistochemical analyses were performed as described in our previous report (Zhang et al., 2021).

2.10. Western blotting analysis

The intestinal tissue homogenate was prepared as described in Section 2.7, but instead of NaCl solution, the Western and IP cell lysis buffer containing inhibitors of phosphatase and proteases was used. The protein expression analysis was carried out in accordance with our previous report (Zhang et al., 2021).

2.11. Statistical analysis

Results were expressed as mean $\pm$ standard error (S.E.), and the OriginLab software (Northampton, USA) was used to determine statistical significance through the one-way ANOVA with Tukey's test ($p < 0.05$). Immunohistochemical and immunofluorescence results were estimated using the Image-Pro Plus 6.0 (Media Cybernetics Inc., Rockville, USA).

3. Results

3.1. Characterization of phytochemicals in the extract

The chromatogram of RM is displayed in Fig. S1, and the detected phytochemical substances are summarized in Table S1. An organic acid and 13 phenolic compounds were detected. The composition was similar with the result reported earlier (Zhang et al., 2021). Gallic acid and quercetin-3-O-rhamnoside were the two predominant phenolic compounds, accounting for 48.78% and 27.54%, respectively, of the total identified phenolic compounds in the extract.

3.2. Macroscopic evaluation

Six pups died. No death was reported in the control group, whereas 3, 2, and 1 pups in the NEC, NEC + L-RM, and NEC + H-RM groups, respectively, died. As shown in Fig. 1A, the survival rate of pups in the control group was 100%, whereas those of pups in the NEC, NEC + L-RM, and NEC + H-RM groups were 70%, 80%, and 90%, respectively. The daily clinical sickness scores of pups are shown in Fig. 1B. The clinical sickness score of pups in the NEC group increased continuously with the induction of the NEC procedure. On the fourth day (Fig. 1C), the clinical sickness score of rat pups in the NEC group was as high as

7.71 ± 0.57 , and the treatment of RM could effectively reduce the clinical sickness score associated with NEC. The clinical sickness scores of pups in the NEC + L-RM and NEC + H-RM groups were 5.46 ± 0.45 and 4.50 ± 0.54 , respectively, showing a distinct decline than that of the NEC group by 29.18% and 41.63% ($p < 0.05$), respectively. However, the clinical sickness scores of rat pups in both treated groups showed no evident difference ($p > 0.05$).

The results of intestinal macroscopic evaluation are shown in Fig. 2. NEC pups had bleeding and ileal swelling, whereas NEC pups administered with RM showed that bleeding and ileal swelling were alleviated and did not have the above symptoms (Fig. 2A and B). As shown in Fig. 2C, the macroscopic assessment score of NEC pups was 4.93 ± 0.20 , which was prominently elevated than that of normal pups ($p < 0.05$). Nevertheless, RM treatment could decrease the macroscopic assessment score. The macroscopic assessment score of pups in the NEC + H-RM group distinctly declined by 41.99% compared with that of the NEC group ($p < 0.05$).

3.3. Histopathological analysis

Intestinal injury was observed by H&E staining. As shown in Fig. 3A, the intestines of rat pups in the control group had normal structure, whereas those in the NEC group had separated villus-core, exfoliated villus epithelium, and lost and even necrotic villi. The histological score of rat pups in the NEC group was 3.17 ± 0.31 , which was dramatically enhanced in comparison with that of rat pups in the control group ($p < 0.05$, Fig. 3B). When NEC pups were administered with RM, the disorder symptoms of villus cell arrangement, villus-core separation, epithelial exfoliation, and villus loss in the intestines were all alleviated. The histological scores of rat pups in NEC + L-RM and NEC + H-RM groups were 2.14 ± 0.26 and 1.50 ± 0.19 , respectively, and were lower than those of pups in the NEC group by 32.49% and 52.68%, respectively (p

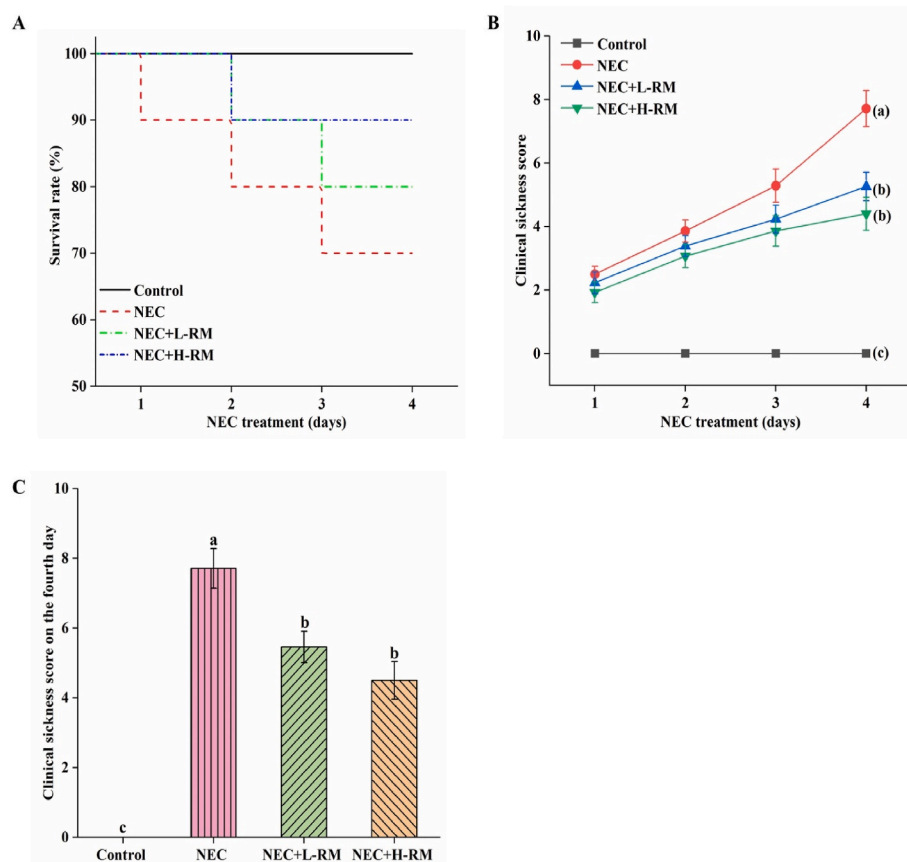


Fig. 1. Effects of the ethanol extract of *R. chinensis* Mill. fruits (RM) on macroscopic symptoms of necrotizing enterocolitis (NEC) pups. (A) Survival rate; (B) clinical sickness score; (C) clinical sickness score on the fourth day. Control: control group, NEC: NEC group (model), NEC + L-RM: NEC + L-RM group (200 mg/kg, ethanol extract), and NEC + H-RM: NEC + H-RM group (400 mg/kg, ethanol extract). All the values are expressed as mean $\pm$ S.E. (Control group: $n = 10$, NEC group: $n = 7$, NEC + L-RM group: $n = 8$, NEC + H-RM $n = 9$). Different letters of the same indicator indicate significant differences ($p < 0.05$).

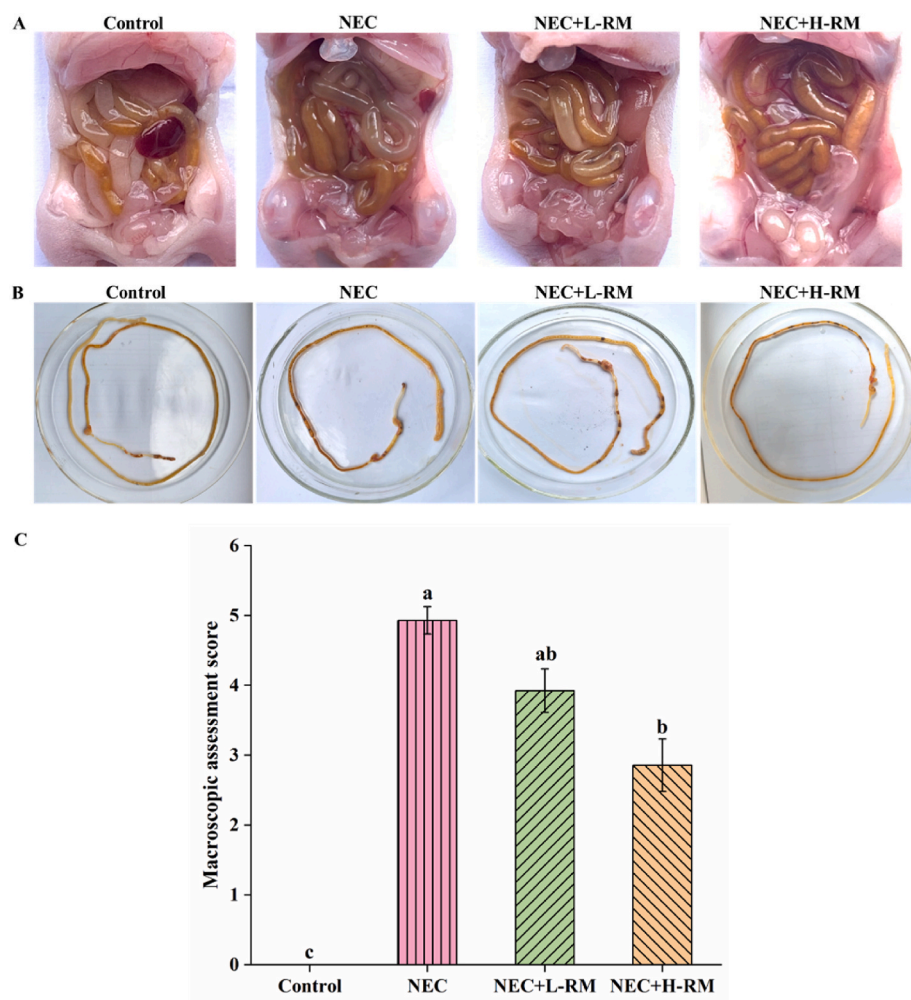


Fig. 2. Effects of the ethanol extract of *R. chinensis* Mill. fruits (RM) on macroscopic evaluation of intestines in necrotizing enterocolitis (NEC) pups. (A) Gross appearance of the intestines; (B) macroscopic appearance of the intestine; (C) macroscopic assessment score. Control: control group, NEC: NEC group (model), NEC + L-RM: NEC + L-RM group (200 mg/kg, ethanol extract), and NEC + H-RM: NEC + H-RM group (400 mg/kg, ethanol extract). All the values are expressed as mean $\pm$ S.E. (Control group: $n = 10$, NEC group: $n = 7$, NEC + L-RM group: $n = 8$, NEC + H-RM: $n = 9$). Different letters of the same indicator indicate significant differences ($p < 0.05$).

< 0.05 , Fig. 3B).

3.4. Analysis of biochemical parameters

The biochemical parameters of intestinal tissues are displayed in Table 3. The intestinal tissue of NEC rat pups had elevated TOS and MDA levels and decreased TAS, SOD, and GSH-Px levels compared with those of the control rat pups ($p < 0.05$). However, in comparison with the NEC group, RM-treated groups had declined TOS and MDA levels and increased TAS level ($p < 0.05$). The SOD and GSH levels in the high-dose group increased evidently ($p < 0.05$). At the same time, the levels of MPO, TNF- α , and IL-6 in the intestinal tissue of NEC rat pups were prominently higher than those in the intestinal tissue of normal rat pups ($p < 0.05$). After RM treatment, the intestinal tissues of both groups had significantly decreased MPO, TNF- α , and IL-6 levels in comparison with those of the NEC group ($p < 0.05$).

3.5. Intestinal barrier function analysis

The expression levels of ZO-1 and Occludin proteins were determined via immunofluorescence assay. As displayed in Fig. 4, the levels of intestinal proteins ZO-1 and Occludin in the control group were high, and their expression levels decreased dramatically in the NEC group ($p < 0.05$). The expression levels of ZO-1 (Fig. 4A and B) and Occludin (Fig. 4C and D) proteins of RM-treated groups were dramatically higher than those of the NEC group ($p < 0.05$).

3.6. Expression of proteins related to oxidative stress, inflammation, and apoptosis

Protein expression levels connected with cellular oxidative stress, inflammation, and apoptosis were determined by immunohistochemistry, immunofluorescence assay, and Western blot analysis (Figs. 5–7, respectively). The levels of Nrf2 and NQO1, the pivotal proteins that participated in oxidative stress, were determined in this study. As shown in Fig. 5A and B, the expression levels of those two proteins in the intestines of NEC pups visibly declined in comparison with those in the intestines of control pups ($p < 0.05$). However, RM-treated groups had significantly enhanced expression levels of these two proteins compared with the NEC group ($p < 0.05$). Immunofluorescence assay was used to evaluate the Nrf2 entering the nucleus. Results are shown in Fig. 5C and D. Compared with that in the intestines of the control group, the nucleus-Nrf2 content in the intestines of the NEC group declined distinctly ($p < 0.05$). After RM administration, the NEC + L-RM and NEC + H-RM groups had increased contents of nucleus-Nrf2, which were higher by about 2.71 and 4.65 times, respectively, than those in the NEC group ($p < 0.05$).

The results of related proteins involved in intestinal inflammation, including TLR4, NF- κ B, p-NF- κ B, and iNOS, are shown in Fig. 6A and B. The levels of TLR4, p-NF- κ B/NF- κ B, and iNOS in the intestines of pups in the NEC group significantly increased compared with those in the intestines of pups in the control group ($p < 0.05$). Compared with those in the intestines of the NEC group, the levels of TLR4, p-NF- κ B/NF- κ B, and iNOS in the intestines of RM-treated groups were visibly declined ($p < 0.05$).

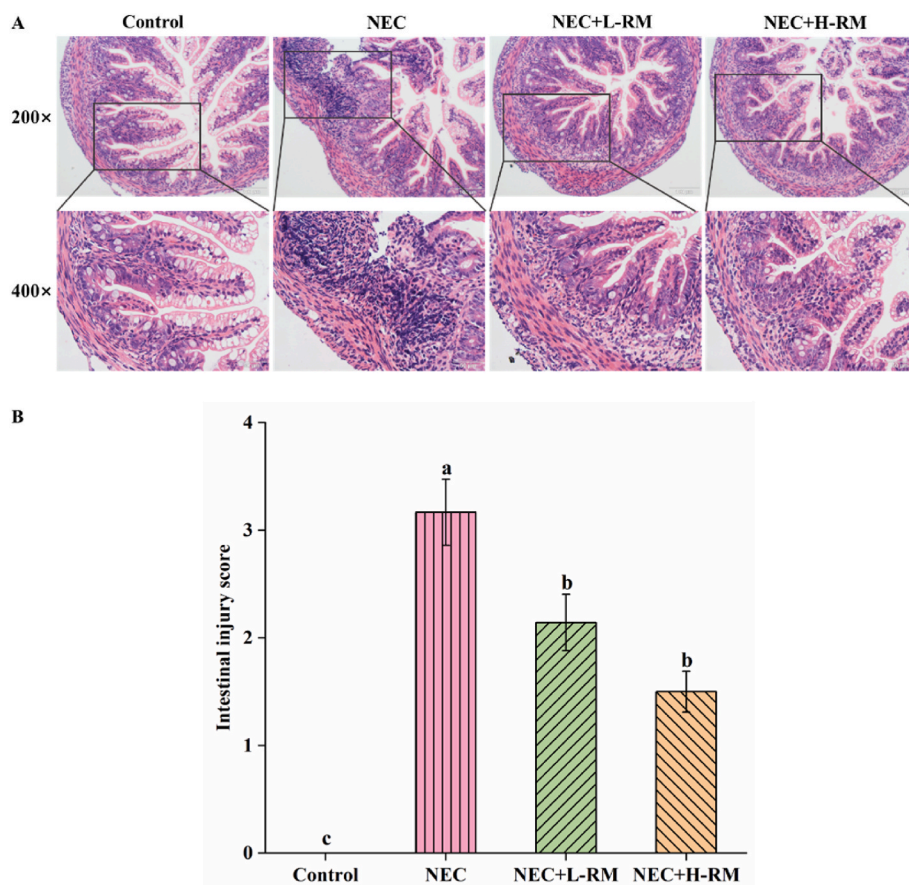


Fig. 3. Effects of the ethanol extract of *R. chinensis* Mill. fruits (RM) on the intestinal injury of necrotizing enterocolitis (NEC) pups. (A) Distal ileum section, 200 × magnification and 400 × magnification, arrows: separation of villus-core; (B) intestinal injury score. Control: control group, NEC: NEC group (model), NEC + L-RM: NEC + L-RM group (200 mg/kg, ethanol extract), and NEC + H-RM: NEC + H-RM group (400 mg/kg, ethanol extract). All the values are expressed as mean ± S.E. (n = 7). Different letters of the same indicator indicate significant differences (p < 0.05).

Table 3
Comparison of biochemical evaluations for each group.

Groups	Control	NEC	NEC + L-RM	NEC + H-RM
TOS (pg/mg protein)	28.68 ± 1.23 ^d	100.92 ± 4.40 ^a	67.49 ± 3.14 ^b	48.45 ± 3.56 ^c
TAS (pg/mg protein)	37.02 ± 1.10 ^a	9.06 ± 0.94 ^d	15.07 ± 1.15 ^c	22.66 ± 1.42 ^b
MDA (nmol/g protein)	2.13 ± 0.38 ^d	14.07 ± 1.13 ^a	10.93 ± 0.64 ^b	6.15 ± 0.73 ^c
SOD (U/mg protein)	110.60 ± 3.00 ^a	60.98 ± 6.01 ^c	73.79 ± 3.25 ^{bc}	83.43 ± 5.98 ^b
GSH-Px (mU/mg protein)	29.00 ± 1.06 ^a	13.14 ± 2.30 ^c	16.32 ± 2.08 ^{bc}	19.57 ± 1.79 ^b
MPO (U/g tissue wet weight)	1.30 ± 0.52 ^d	6.96 ± 0.15 ^a	5.15 ± 0.60 ^b	3.23 ± 0.36 ^c
IL-6 (pg/mg protein)	32.08 ± 2.64 ^d	112.85 ± 7.94 ^a	80.56 ± 5.65 ^b	58.29 ± 4.96 ^c
TNF-α (pg/mg protein)	4.01 ± 0.33 ^d	12.46 ± 0.79 ^a	9.60 ± 0.46 ^b	7.15 ± 0.62 ^c

TOS: Total oxidant status; TAS: Total antioxidant status; MDA: Malondialdehyde; MPO: Myeloperoxidase; SOD: Superoxide dismutase; GSH-Px: Glutathione peroxidase; IL-6: Interleukin-6; TNF-α: Tumor necrosis factor-α; NEC: Necrotizing enterocolitis; RM: Ethanol extract of *R. chinensis* Mill. fruits. Control: control group, NEC: NEC group (model), NEC + L-RM: NEC + L-RM group (200 mg/kg, ethanol extract), and NEC + H-RM: NEC + H-RM group (400 mg/kg, ethanol extract). All the values are expressed as mean ± SE (Control group: n = 10, NEC group: n = 7, NEC + L-RM group: n = 8, NEC + H-RM group: n = 9). Different letters of the same indicator indicate significant differences (p < 0.05).

0.05). Furthermore, the intestinal TLR4 and iNOS levels in the NEC + H-RM group were close to those in the control group (p > 0.05). In addition, the expression of TLR4 in the intestine was detected by

immunohistochemistry. Results are shown in Fig. 6C and D. The expression level of TLR4 in the intestines of the NEC group were significantly augmented in comparison with that in the intestines of the control group (p < 0.05). However, the TLR4 expression levels of RM-treated groups declined remarkably compared with that of the NEC group (p < 0.05). The TLR4 expression of the NEC + H-RM group had no evident difference compared with that of the control group (p > 0.05). p-NF-κB entering the nucleus could stimulate the production of downstream proinflammatory mediators, and the expression of p-NF-κB in the nucleus was determined by immunohistochemistry. As displayed in Fig. 6E and F, the NEC group had a high level of nucleus-p-NF-κB level compared with the control group (p < 0.05). The expression levels of nucleus-p-NF-κB in RM-treated groups declined remarkably in comparison with that in the NEC group (p < 0.05).

The TUNEL assay was used to analyze cell apoptosis. As displayed in Fig. 7A and B, few TUNEL-positive signals were observed in the intestines of control pups, whereas more TUNEL-positive signals appeared in the intestines of NEC pups (p < 0.05). TUNEL-positive signals in the pups' intestines of NEC + L-RM and NEC + H-RM groups declined significantly (p < 0.05). Moreover, the expression levels of three pivotal proteins involved in cell apoptosis, namely, cleaved Caspase-3, Bax, and Bcl-2, were determined. Results are displayed in Fig. 7C and D. Compared with control pups, NEC pups had elevated expression levels of cleaved Caspase-3 and Bax and declined level of Bcl-2 (p < 0.05). Furthermore, the intestines of pups in RM-treated groups had evidently decreased cleaved Caspase-3 and Bax levels and significantly increased Bcl-2 level compared with those in the NEC group (p < 0.05). Moreover, the expression levels of the three proteins in the rat pups of the NEC + H-RM group were similar to those in the rat pups of the control group (p > 0.05).

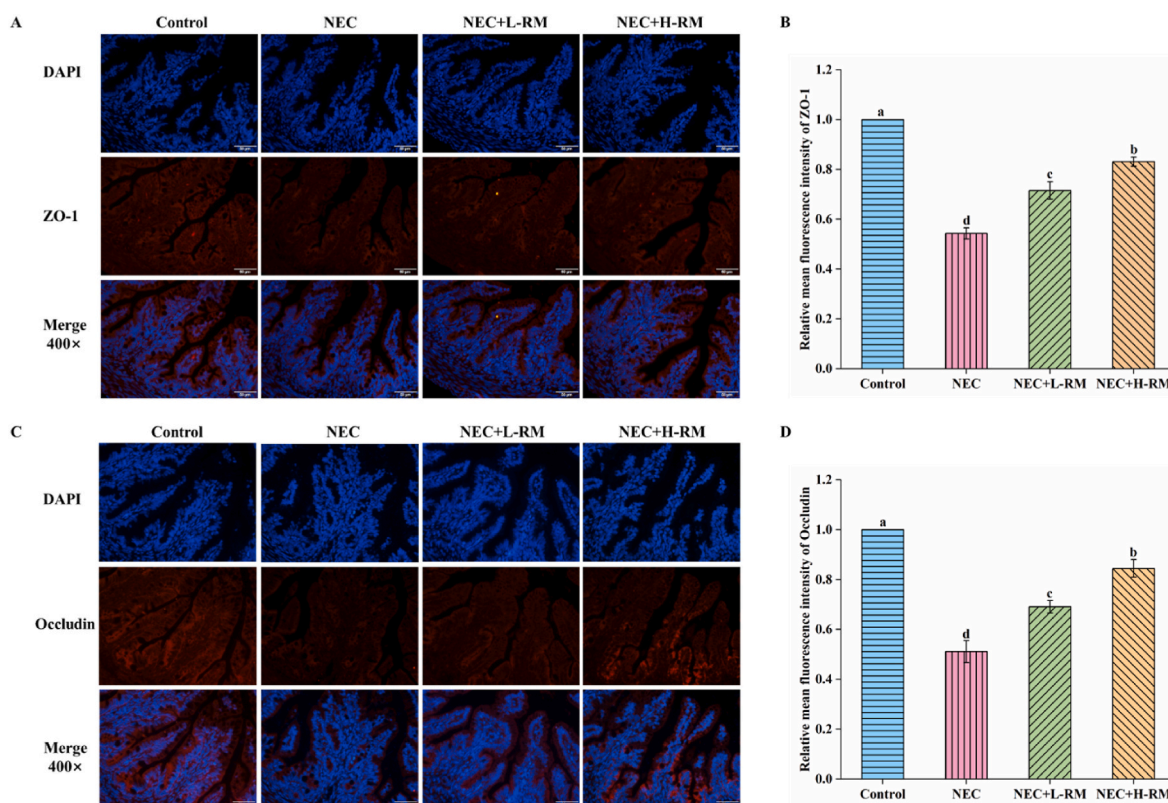


Fig. 4. Effects of the ethanol extract of *R. chinensis* Mill. fruits (RM) on the intestinal barrier of necrotizing enterocolitis (NEC) pups. (A) Zonula Occludens 1 (ZO-1) immunostaining and 4,6-diamino-2-phenylindole (DAPI) in intestinal sections, 400 × magnification. The immunofluorescence results of ZO-1: the red fluorescence part is the protein of ZO-1. (B) The relative protein expression of ZO-1. The relative protein expression in immunohistochemistry was quantified by using Image-Pro Plus software. The quantitative result is expressed in terms of density mean and the above proteins were quantified by normalizing with the control group. (C) Occludin immunostaining and DAPI in intestinal sections, 400 × magnification. The immunofluorescence results of Occludin: the red fluorescence part is the protein of Occludin. (D) The relative protein expression of Occludin. The relative protein expression in immunohistochemistry was quantified by using Image-Pro Plus software. The quantitative result is expressed in terms of density mean and the above proteins were quantified by normalizing with the control group. Control: control group, NEC: NEC group (model), NEC + L-RM: NEC + L-RM group (200 mg/kg, ethanol extract), and NEC + H-RM: NEC + H-RM group (400 mg/kg, ethanol extract). All the values are expressed as mean ± S.E. ($n = 7$). Different letters of the same indicator indicate significant differences ($p < 0.05$). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

4. Discussions

RM treatment has a good improvement effect on NEC. Clinical sickness score, mortality, histopathological score, and intestinal macroscopic conditions are all visibly improved. Results of several biochemical indicators, immunohistochemical, immunofluorescence, and Western blot assays further confirmed that RM treatment can visibly improve intestinal mucosal integrity, oxidative stress, inflammation, and cell apoptosis, thereby preventing NEC occurrence.

Intestinal barrier damage is the basis of the pathophysiology of NEC (Blair et al., 2006). Intestinal structural disruption leads to increased intestinal epithelial barrier permeability, bacterial translocation, and inflammation aggravation, and these changes also cause epithelial cell edema and apoptosis (Nowicki, 2005). According to the literature, the destruction of tight junction (TJ) proteins is an important feature of NEC (Xiao et al., 2018). The TJ protein is an intercellular adhesion structure composed of protein complexes on the cell membrane and plays a crucial role in keeping the integrity of the intestinal epithelial barrier (Blackwood et al., 2017). ZO-1 and Occludin are important TJ proteins in intestinal epithelial cells. At present, improving the expression of TJ protein to maintain the integrity of the intestinal barrier is one of the crucial ways for many natural products to prevent NEC. For example, the soybean-derived recombinant human epidermal growth factor weakens the dissociation of ZO-1 protein, and Lactadherin improves the localization of Occludin in the intestine, thereby prominently alleviating the symptoms of NEC (Isani et al., 2018; Shen et al., 2019). In addition, a

previous study found that RM treatment improves the integrity of the intestinal barrier by increasing the levels of TJ proteins Occludin and ZO-1, thereby alleviating DSS-induced colitis (Zhang et al., 2021). In the current study, the expression levels of ZO-1 and Occludin of rat pups in the NEC group dropped dramatically, indicating that the symptoms of NEC are severe. However, RM treatment can visibly alleviate the decrease in ZO-1 and Occludin expression levels. The above results indicate that RM may also act on TJ proteins to maintain the integrity of the intestinal barrier structure for improved permeability of the intestine, thereby alleviating the symptoms of NEC.

Oxidative stress plays a vital role in the occurrence and development of NEC. The immature and unbalanced antioxidant system in newborns is prone to damage, which produces oxidative stress (Cakir et al., 2018). In addition, the intestinal ischemia of pups caused by the NEC procedure can lead to the production of reactive oxygen species (ROS). ROS disrupts redox balance, cell structure, and intestinal integrity (Cakir et al., 2018; Wang et al., 2019). Therefore, high levels of ROS and immature antioxidant defense mechanisms are the main factors for the oxidative destruction of intestinal cells. TAS and TOS are important indicators used to assess the overall changes in the oxidation state of intestinal tissue (Erel, 2005). NEC pups are observed with elevated TOS and reduced TAS, indicating that the redox state of the intestine is disrupted by the NEC procedure. After RM (200 or 400 mg/kg) administration, the TOS level is dramatically declined, and the TAS level is prominently enhanced in the intestinal tissue of pups. According to a previous report, the *R. coriaria* water extract (800 mg/kg) shows antioxidant activity,

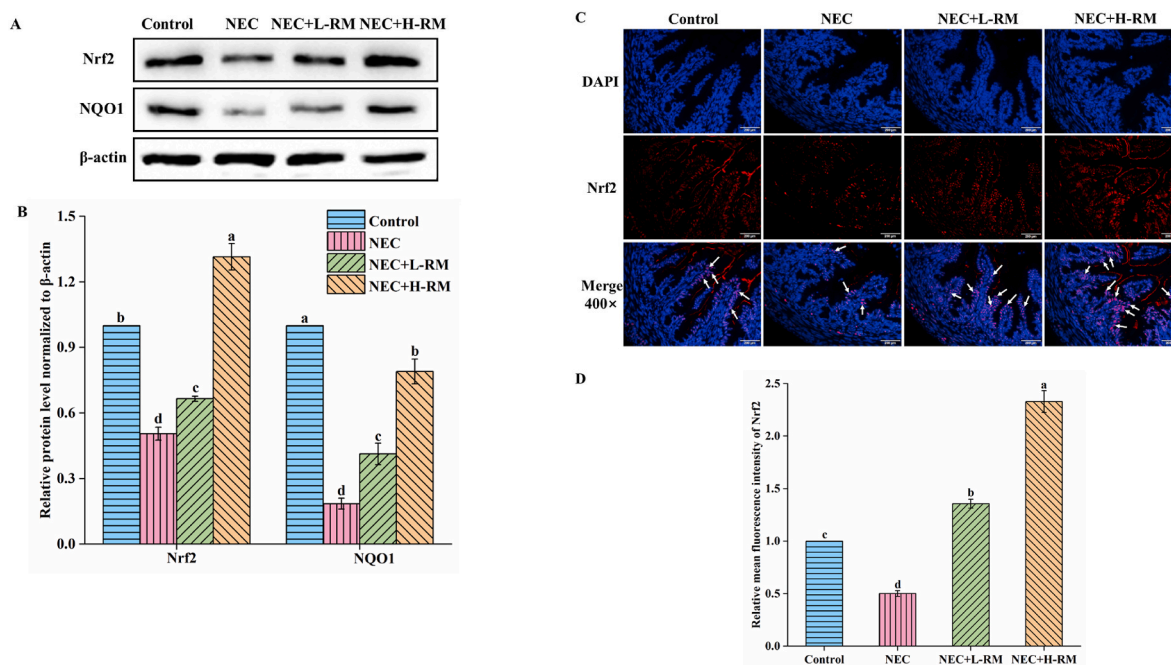


Fig. 5. Effects of the ethanol extract of *R. chinensis* Mill. fruits (RM) on the expression of key proteins in the oxidative stress pathway in the intestines of necrotizing enterocolitis (NEC) pups. Samples were obtained from the intestinal homogenate of the experimental pups. (A) Western blotting was used to detect the protein levels of Nuclear factor erythroid 2-related factor 2 (Nrf2) and NAD(P)H-quinone oxidoreductase-1 (NQO1); (B) the control group normalization method was used to quantify the relative expression of the above proteins and compared with β-actin. (C) Nrf2 immunofluorescence, 400 × magnification. The results of immunofluorescence: the red part is the Nrf2 protein; the blue is the nucleus; the overlap of red and blue fluorescence indicates that the Nrf2 protein enters the nucleus. (D) The relative protein expression of Nrf2 entering into the nucleus. The relative protein expression in immunofluorescence was quantified by using Image-Pro Plus software. The quantitative result is expressed in terms of density mean and the above proteins were quantified by normalizing with the control group. Control: control group, NEC: NEC group (model), NEC + L-RM: NEC + L-RM group (200 mg/kg, ethanol extract), and NEC + H-RM: NEC + H-RM group (400 mg/kg, ethanol extract). All the values are expressed as mean ± S.E. ($n = 7$). Different letters of the same indicator indicate significant differences ($p < 0.05$). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

which dramatically decreases the TOS level and increases the TAS level (Isik et al., 2019). In the current work, the dose of RM is lower than that of the *R. coriaria* water extract, indicating that RM may have better antioxidant properties than the water extract of *R. coriaria*. Therefore, RM may also exert antioxidant effects by increasing the TAS level and reducing the TOS level, thereby alleviating the changes in the oxidative stress state of pups caused by the NEC procedure. At the same time, ROS, a kind of powerful oxidant, can destroy cell structure and cause cell necrosis through membrane lipid peroxidation (Tayman et al., 2016). MDA is the end-product of membrane lipid peroxidation and partly reflects the degree of oxidative stress (Cigsar et al., 2020; Zhang et al., 2019). Some antioxidant enzymes in the immature intestine of newborns can resist NEC. For example, high levels of SOD have a protective effect on intestinal ischemia–reperfusion injury, and GSH-Px can resist the toxic metabolites of oxidative stress (Chan et al., 2002; Özdemir et al., 2010). However, the loss of intestinal integrity leads to weakened newborns' antioxidant enzyme capacity (Yurttutan et al., 2013). Currently, the literatures reported that bioactive ingredients in traditionally used medicinal plants can prevent NEC by improving enzymes related to oxidative stress. For example, the *Nigella sativa* oil treatment dramatically increases the antioxidant enzymes in intestinal tissues, such as SOD and GSH-Px, to reduce the MDA content, thereby reducing NEC-related damage (Tayman et al., 2012). In the present study, an evident augmentation in MDA concentration and distinct decreases in the SOD and GSH-Px activities of rat pups in the NEC group indicate that the NEC procedure may promote the progress of oxidative stress and destroy the antioxidant capacity. However, RM reduces the production of MDA and promotes the activities of SOD and GSH-Px to alleviate the damage of ROS. Moreover, Nrf2 is considered to be the main regulator of redox homeostasis (Li et al., 2021). Under steady-state conditions, Nrf2 binds to Keap1 in the cytoplasm. When subjected to oxidative or

electrophilic stress, Nrf2 dissociates from Keap1, enters the nucleus, and combines with ARE to promote the expression of some antioxidant-active proteins (Yang et al., 2018). NQO1, the marker protein of the downstream antioxidant activity of Nrf2, can maintain the redox state of cells, reduce ROS production, and resist damage caused by oxidative stress (Ross and Siegel, 2017). Previous studies proved that the Nrf2 pathway is a key component of the antioxidant mechanism in the regulation of intestinal disease. For example, RM inhibits oxidative stress by up-regulating the nucleus–Nrf2 and NQO1 levels in the DSS-induced colitis experiment in mice, thereby reducing the severity of colitis (Zhang et al., 2021). In the present work, the levels of nucleus–Nrf2 and NQO1 are remarkably reduced when rat pups are treated with the NEC procedure and increase when rat pups are administered with RM. Promoting nucleus–Nrf2 and NQO1 expression levels by using RM may be one of the underlying mechanisms to eliminate excessive ROS for reduced state of oxidative stress in the intestine, thereby reducing the NEC damage.

The abnormal activation of the intestinal immune system is also considered as a significant part of the pathogenesis of NEC (Lim et al., 2015). Previous studies showed that the activation of proinflammatory cytokines (such as TNF-α and IL-6) is a vital feature of the innate immune response of NEC (Niño et al., 2016). TNF-α is a proinflammatory cytokine produced by immune cells, aggravates mucosal damage, changes intestinal permeability, and induces bacterial translocation in NEC (Hunter and De Plaen, 2014; Nakame et al., 2016). TNF-α can also accelerate the production of other inflammatory cytokines, such as IL-6 (Hunter and De Plaen, 2014; Szliszka and Krol, 2011). The IL-6 level may reflect bacterial invasion and intestinal damage caused by NEC (Xiao et al., 2018). Furthermore, the intestinal microcirculation of newborns is not fully mature and is prone to hypoxia and ischemia (Lim et al., 2015). Hypoxia and ischemia lead to the migration and activation

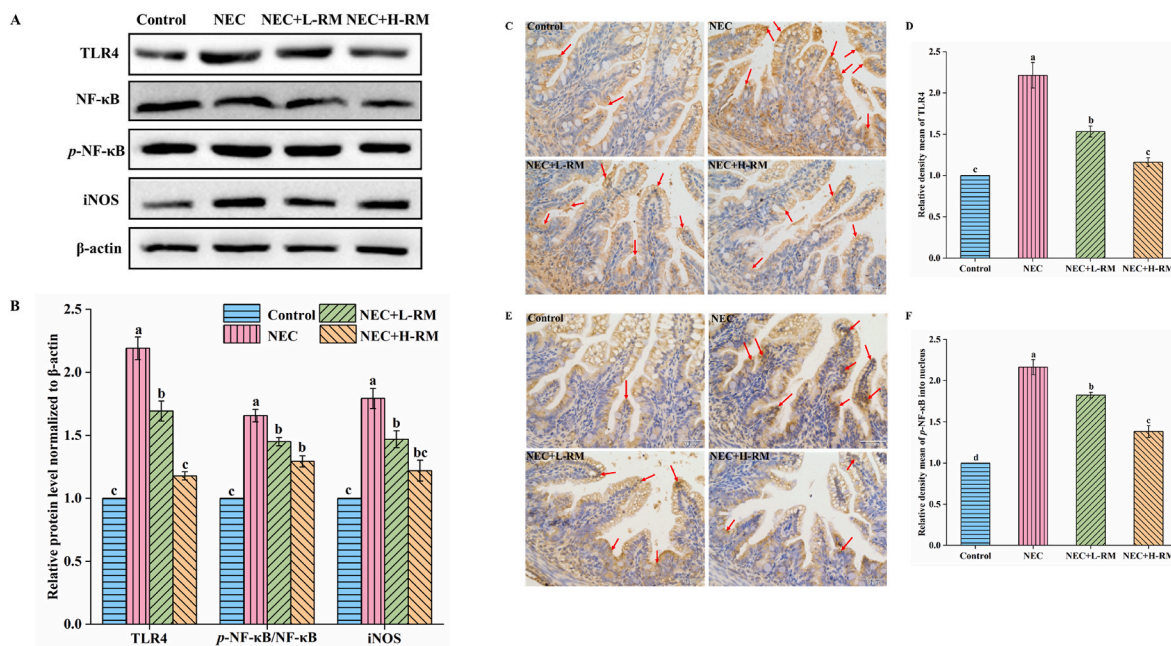


Fig. 6. Effects of the ethanol extract of *R. chinensis* Mill. fruits (RM) on the expression of key proteins in the inflammation pathway in the intestines of necrotizing enterocolitis (NEC) pups. Samples were obtained from the intestinal homogenate of the experimental pups. (A) Western blotting was used to detect the protein levels of Toll-like receptor 4 (TLR4), Nuclear factor kappaB (NF-κB), Phosphorylated-Nuclear factor kappaB (p-NF-κB), and Inducible nitric oxide synthase (iNOS); (B) the control group normalization method was used to quantify the relative expression of the above proteins and compared with β-actin. (C) TLR4 immunohistochemistry, 400 × magnification. The results of immunohistochemistry: the yellow part is the TLR4 protein; (D) The relative protein expression of TLR4. The TLR4 expression in immunohistochemistry was quantified by using Image-Pro Plus software. The quantitative result is expressed in terms of density mean and the above proteins were quantified by normalizing with the control group. (E) p-NF-κB immunohistochemistry, 400 × magnification. The results of immunohistochemistry: the yellow part is the p-NF-κB protein; the blue is the nucleus; the overlap of yellow and blue fluorescence indicates that the p-NF-κB protein enters the nucleus to induce inflammation. (F) The relative protein expression of p-NF-κB entering into the nucleus. The p-NF-κB expression in immunohistochemistry was quantified by using Image-Pro Plus software. The quantitative result is expressed in terms of density mean and the above proteins were quantified by normalizing with the control group. Control: control group, NEC: NEC group (model), NEC + L-RM: NEC + L-RM group (200 mg/kg, ethanol extract), and NEC + H-RM: NEC + H-RM group (400 mg/kg, ethanol extract). All the values are expressed as mean ± S.E. (n = 7). Different letters of the same indicator indicate significant differences (p < 0.05). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

of neutrophils, releasing cytokines and ROS, and aggravating inflammation (Hunter and De Plaen, 2014; Neu and Walker, 2011). MPO is related to neutrophils and indirectly evaluates the migration and infiltration of neutrophils in the intestine. In the current study, the levels of MPO, IL-6, and TNF-α in RM-treated pups have significantly declined compared with those in NEC pups, indicating that RM alleviates NEC-induced intestinal inflammation in pups. The water extract of *R. coriaria* can also significantly inhibit the increases in IL-6 and TNF-α levels caused by NEC (Isik et al., 2019). However, the dose of *R. coriaria* water extract (800 mg/kg) is higher than that of RM (200 or 400 mg/kg), indicating that the latter may have a stronger anti-inflammatory effect than the former. Yakut et al. also reported that *Foeniculum vulgare* effectively prevents NEC by reducing the levels of MPO, IL-6, and TNF-α in intestinal tissues (Yakut et al., 2020). This finding is consistent with the results of the present study. In addition, according to previous literature, TLR4 knockout mice cannot develop experimental NEC, and the expression of TLR4 in the intestines occurs before tissue damage (Jilling et al., 2006; Liu et al., 2009), indicating that the expression of TLR4 may be a key step in the occurrence and development of NEC. Other studies also pointed out that the TLR4 overexpression results in decreased cell proliferation, increased apoptosis, lowered barrier integrity, and augmented bacterial translocation in the intestines, thereby causing inflammation (Neal et al., 2012; Neu and Walker, 2011). From the perspective of inflammation, activating TLR4 stimulates the release and activation of NF-κB and makes activated NF-κB migrate into the nucleus to stimulate the production of inflammatory mediators (iNOS), thereby aggravating the NEC (Liu et al., 2018). Jing et al. reported that berberine can down-regulate the expression of TLR4 to inhibit the production of inflammatory

mediators, thereby reducing the incidence of NEC (Jing et al., 2018). Yang et al. also reported that human breast milk oligosaccharides can reduce NEC damage in rats by inhibiting the ileal tissue expression of TLR4 (He-Yang et al., 2020). In the current research, the expression levels of TLR4, p-NF-κB, and iNOS in the intestines dramatically increased in rat pups under NEC procedure, indicating that the NEC procedure promotes inflammation. Nevertheless, such elevation is visibly suppressed by RM treatment. These results manifested that RM may alleviate intestinal inflammation by down-regulating the expression levels of the TLR4, p-NF-κB/NF-κB and iNOS proteins and suppressing nuclear translocation of p-NF-κB to prevent NEC development.

Newborns are prone to inappropriate apoptotic reactions due to immature intestinal development (Hunter and De Plaen, 2014). A previous study also found that apoptosis is the earliest histopathological change in intestinal tissues in the development of NEC and is a key step in the pathogenesis of NEC (Jilling et al., 2004). In this present study, TUNEL staining is used to observe cell apoptosis intuitively. The cell apoptosis in intestinal tissues increases distinctly under the NEC procedure. This phenomenon further promotes the development of NEC in pups. Under RM treatment, apoptosis dramatically declines, indicating that RM may reduce the incidence and severity of NEC by decreasing cell apoptosis. In the development of NEC, oxidative stress and inflammation can cause evident mitochondrial disorders and activate mitochondria-mediated apoptosis, further leading to the death of intestinal epithelial cells in NEC pups (Han et al., 2016). Therefore, the regulation of RM on apoptosis driven by the mitochondrial pathway is further studied. Bcl-2 family proteins, as key promoters of the mitochondrial apoptosis pathway, can regulate the permeability of the mitochondrial outer membrane (Zhou et al., 2018). The antiapoptotic

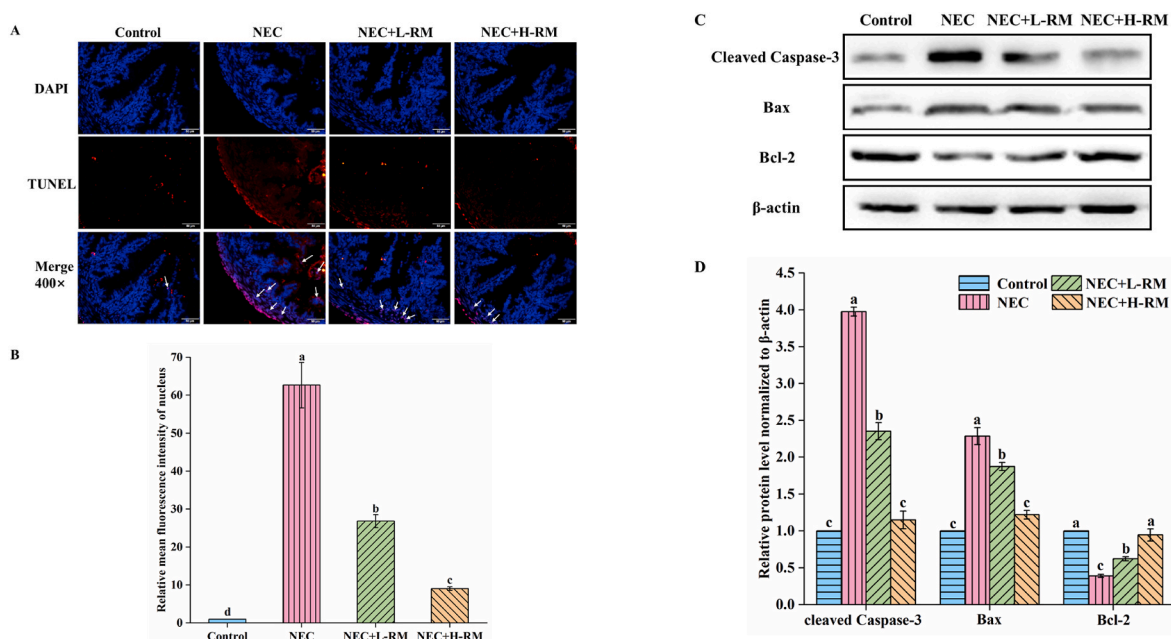


Fig. 7. Effects of the ethanol extract of *R. chinensis* Mill. fruits (RM) on apoptosis in the intestine of necrotizing enterocolitis (NEC) pups. (A) Terminal deoxynucleotidyl transferase (TdT)-mediated dUTP nick end labeling (TUNEL) assay of distal ileum section, 400 × magnification. The results of the TUNEL assay: the red fluorescence part is the broken DNA; the blue fluorescence is the nucleus, and the overlap of red and blue fluorescence indicates the cell apoptosis. (B) The relative mean fluorescence intensity of nucleus. The relative nucleus expression in immunofluorescence was quantified by using Image-Pro Plus software and was quantified by normalizing with the control group. (C) Western blotting was used to detect the protein levels of cleaved cysteinyl aspartate specific proteinase-3 (cleaved Caspase-3), Bcl-2-associated X (Bax), and B-cell lymphoma-2 (Bcl-2), (D) the control group normalization method was used to quantify the relative expression of the above proteins and compared with β-actin. Control: control group, NEC: NEC group (model), NEC + L-RM: NEC + L-RM group (200 mg/kg, ethanol extract), and NEC + H-RM: NEC + H-RM group (400 mg/kg, ethanol extract). All the values are expressed as mean ± S.E. ($n = 7$). Different letters of the same indicator indicate significant differences ($p < 0.05$). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

protein Bcl-2 and the proapoptotic protein Bax are typical Bcl-2 family proteins (Yang et al., 2020; Zhou et al., 2018). Bcl-2 inhibits the release of cytochrome C and other proapoptotic factors by regulating the permeability of the outer mitochondrial membrane, thereby maintaining cell viability; whereas Bax promotes the release of cytochrome C from mitochondria into the cytoplasm, thereby activating the caspase cascade (Eimani et al., 2014). Caspase-3 is also an important protein involved in this apoptosis pathway, and activated Caspase-3 (cleaved Caspase-3) directly leads to cell apoptosis (Brentnall et al., 2013). A previous study reported that milk polar lipids inhibited mitochondria-mediated intestinal cell apoptosis, thereby alleviating the symptoms of the NEC rat (Yang et al., 2020). In the current study, NEC pups have increased cleaved Caspase-3 and Bax expression levels and decreased Bcl-2 expression compared with control pups, indicating that severe mitochondria-mediated intestinal cell apoptosis occurs in NEC pups. However, RM prominently inhibits the expression levels of cleaved Caspase-3 and Bax and promotes the expression of Bcl-2 to alleviate NEC. Notably, the expression levels of these three proteins of pups in the NEC + H-RM group are similar to those of pups in the control group, suggesting that the high dose of RM shows effective inhibition against mitochondria-mediated apoptosis. These findings indicated that regulating the expression of key proteins in the mitochondrial apoptosis pathway to reduce intestinal tissue cell apoptosis may be also one of the underlying mechanisms of RM to prevent NEC in rat pups.

According to the result of UHPLC-HRMS/MS analysis, phenolic compounds are the main substances in the extract. Quantitative results show that gallic acid and quercetin-3-O-rhamnoside are the most abundant phenolic compounds, accounting for 48.78% and 27.54%, respectively, of the total identified phenolic compounds in the extract. A nonphenolic substance, citric acid, which is an organic acid common in many plants, is also observed. Thus far, a report showing that this organic acid can prevent NEC is not available. However, some previous

studies proved that phenolic-rich extracts or phenolic compounds, such as sesamol, quercetin, and phenolic-rich extract of *R. coriaria*, from different plants can effectively prevent NEC in rat pups (Cigsar et al., 2020; Isik et al., 2019; Ozcan et al., 2018). In the present work, phenolic compounds may also prevent NEC in rat pups through a variety of mechanisms. According to the available literature about phenolic compounds preventing NEC, the preventive effect may substantially depend on the phenolic compositions or phenolic structures. Further studies on the structure-activity relationship of phenolic substances to prevent NEC and the interactions between phenolic substances are needed to understand and apply certain phenolic substances to prevent NEC.

5. Conclusions

This study showed that RM particularly a high dose (NEC + H-RM, 400 mg extract/kg body weight) had a strong preventive effect against NEC. The extract maintained the integrity of the intestinal barrier by increasing the levels of TJ proteins (i.e., ZO-1 and Occludin) and inhibited oxidative stress, inflammation, and apoptosis by regulating Nrf2, TLR4/NF-κB, and Caspase-3 signaling pathways (Fig. 8). Therefore, *R. chinensis* fruits, as a traditional herbal medicine, could effectively prevent NEC in rat pups and might provide a new therapy for the prevention and/or alleviation of NEC.

CRedit authorship contribution statement

Yi Zhang: Investigation, Data curation, Writing – original draft, All authors have read and approved the final version of the manuscript. **Ou Wang:** Methodology, All authors have read and approved the final version of the manuscript. **Hongying Mi:** Writing – review & editing. **Junjie Yi:** Funding acquisition. **Shengbao Cai:** Funding acquisition, Conceptualization, Supervision, All authors have read and approved the

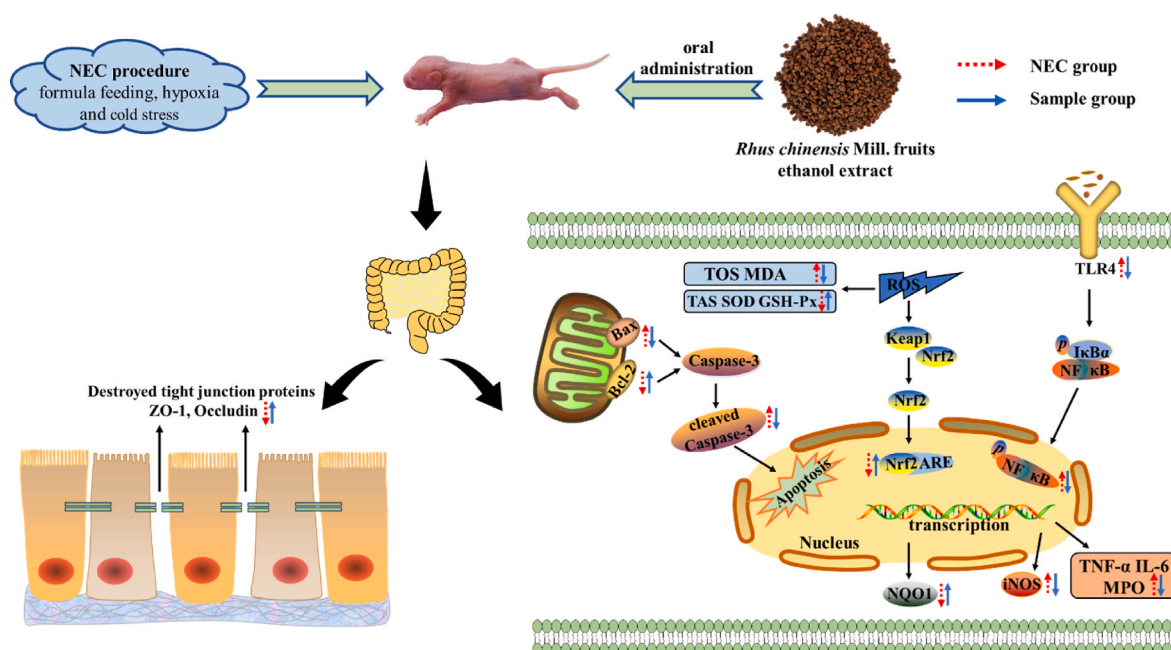


Fig. 8. Ethanol extract of *R. chinensis* Mill. fruits (RM) improve necrotizing enterocolitis (NEC) through oxidative stress, inflammation and apoptosis signaling pathway. ARE: Antioxidant response element; Bax: Bcl-2-associated X; Bcl-2: B-cell lymphoma-2; cleaved Caspase-3: cleaved cysteinyl aspartate specific proteinase-3; GSH-Px: Glutathione peroxidase; IL-6: Interleukin-6; iNOS: Inducible nitric oxide synthase; Keap1: Kelch like-ECH-associated protein 1; MDA: Malondialdehyde; MPO: Myeloperoxidase; NQO1: NAD(P)H-quinone oxidoreductase-1; NEC Necrotizing enterocolitis; Nrf2: Nuclear factor erythroid 2-related factor 2; NF-κB: Nuclear factor kappaB; p-NF-κB: Phosphorylated-Nuclear factor kappaB; SOD: Superoxide dismutase; TAS: Total antioxidant status; TLR4: Toll-like receptor 4; TNF-α: Tumor necrosis factor-α; TOS: Total oxidant status; ZO-1: Zonula Occludens 1.

final version of the manuscript.

Declaration of competing interest

The authors declare no conflicts of interest.

Acknowledgments

The present work was financially supported by the National Natural Science Foundation of China (Grant No. 31960477), Major Science and Technology Project of the Science and Technology Department of Yunnan Province (Grant Nos. 202102AE090050 and 202102AE090025), Science and Technology Project of Yunnan Province (Grant Nos. 202002AE320006-01-03 and 2019ZG001-4-1).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2022.115103>.

Abbreviations

ARE	Antioxidant response element
Bax	Bcl-2-associated X
Bcl-2	B-cell lymphoma-2
Cleaved Caspase-3	Cleaved cysteinyl aspartate specific proteinase-3
DAPI	4,6-Diamino-2-phenylindole
GSH-Px	Glutathione peroxidase
H&E	Hematoxylin–eosin
IL-6	Interleukin-6
iNOS	Inducible nitric oxide synthase
Keap1	Kelch like-ECH-associated protein 1
MDA	Malondialdehyde
MPO	Myeloperoxidase
NQO1	NAD(P)H-quinone oxidoreductase-1

NEC	Necrotizing enterocolitis
Nrf2	Nuclear factor erythroid 2-related factor 2
NF-Kb	Nuclear factor kappaB
p-NF-κB	Phosphorylated-Nuclear factor kappaB
ROS	Reactive oxygen species
RM	Ethanol extract of <i>R. chinensis</i> fruits
S.E.	Standard error
SOD	Superoxide dismutase
TAS	Total antioxidant status
TLR4	Toll-like receptor 4
TNF-α	Tumor necrosis factor-α
TOS	Total oxidant status
TJ	Tight junction
TUNEL	Terminal deoxynucleotidyl transferase (TdT)–mediated dUTP nick end labeling
UHPLC	Ultrahigh-pressure liquid chromatography
ZO-1	Zonula Occludens 1

References

- Blackwood, B.P., Yuan, C.Y., Wood, D.R., Nicolas, J.D., Grothaus, J.S., Hunter, C.J., 2017. Probiotic *Lactobacillus* species strengthen intestinal barrier function and tight junction integrity in experimental necrotizing enterocolitis. *J. Probiotics Health* 5 (1), 159. <https://doi.org/10.4172/2329-8901.1000159>.
- Blair, S.A., Kane, S.V., Clayburgh, D.R., Turner, J.R., 2006. Epithelial myosin light chain kinase expression and activity are upregulated in inflammatory bowel disease. *Lab. Invest.* 86, 191–201. <https://doi.org/10.1038/labinvest.3700373>.
- Bose, S.K., Dewanjee, S., Gupta, A.S., Samanta, K.C., Kundu, M., Mandal, S.C., 2008. *In vivo* evaluation of antidiarrhoeal activity of *Rhus semialata* fruit extract in rats. *Afr. J. Tradit., Complement. Altern. Med.* 5 (1), 97–102. <https://doi.org/10.4314/ajtcam.v5i1.31262>.
- Brentnall, M., Rodriguez-Menocal, L., De Guevara, R.L., Cepero, E., Boise, L.H., 2013. Caspase-9, caspase-3 and caspase-7 have distinct roles during intrinsic apoptosis. *BMC Cell Biol.* 14, 32. <https://doi.org/10.1186/1471-2121-14-32>.
- Cakir, U., Tayman, C., Serkan, U., Yakut, H.I., Cakir, E., Ates, U., Koyuncu, I., Karaogul, E., 2018. *Ginger (Zingiber officinale Roscoe)* for the treatment and prevention of necrotizing enterocolitis. *J. Ethnopharmacol.* 225, 297–308. <https://doi.org/10.1016/j.jep.2018.07.009>.
- Chan, K.L., Hui, C.W.C., Chan, K.W., Fung, P.C.W., Wo, J.Y.H., Tipoe, G., Tam, P.K.H., 2002. Revisiting ischemia and reperfusion injury as a possible cause of necrotizing

- enterocolitis: role of nitric oxide and superoxide dismutase. *J. Pediatr. Surg.* 37 (6), 828–834. <https://doi.org/10.1053/jpsu.2002.32882>.
- Chen, J.-C., Ho, T.-Y., Chang, Y.-S., Wu, S.-L., Hsiang, C.-Y., 2006. Anti-diarrheal effect of *Galla Chinensis* on the *Escherichia coli* heat-labile enterotoxin and ganglioside interaction. *J. Ethnopharmacol.* 103 (3), 385–391. <https://doi.org/10.1016/j.jep.2005.08.036>.
- Cigarsar, E.B., Karadag, C.A., Tanik, C., Aydin, A.F., Dokucu, A.I., 2020. The protective effects of sesamol in a neonatal rat model of necrotizing enterocolitis. *J. Matern. – Fetal Neonatal Med.* 33 (6), 889–894. <https://doi.org/10.1080/14767058.2018.1506759>.
- Djakpo, O., Yao, W., 2010. *Rhus chinensis* and *Galla Chinensis*—folklore to modern evidence. *Phytother. Res.* 24 (12), 1739–1747. <https://doi.org/10.1002/ptr.3215>.
- Drucker, N.A., Jensen, A.R., Ferkowicz, M., Markel, T.A., 2018. Hydrogen sulfide provides intestinal protection during a murine model of experimental necrotizing enterocolitis. *J. Pediatr. Surg.* 53 (9), 1692–1698. <https://doi.org/10.1016/j.jpedsurg.2017.12.003>.
- Eimani, B.G., Sanati, M.H., Houshmand, M., Ateai, M., Akbarian, F., Shakhssalim, N., 2014. Expression and prognostic significance of bcl-2 and bax in the progression and clinical outcome of transitional bladder cell carcinoma. *Cell J* 15 (4), 356–363.
- Erel, O., 2005. A new automated colorimetric method for measuring total oxidant status. *Clin. Biochem.* 38 (12), 1103–1111. <https://doi.org/10.1016/j.clinbiochem.2005.08.008>.
- Gong, H., Yuan, Q., Pang, J., Li, T., Li, J., Zhan, B., Chang, R., Mao, X., 2020. Dietary milk fat globule membrane restores decreased intestinal mucosal barrier development and alterations of intestinal flora in infant-formula-fed rat pups. *Mol. Nutr. Food Res.* 64 (21), 2000232. <https://doi.org/10.1002/mnfr.202000232>.
- Han, J., Zhao, Q., Basmadjian, C., Désaubry, L., Theiss, A.L., 2016. Flavaglines ameliorate experimental colitis and protect against intestinal epithelial cell apoptosis and mitochondrial dysfunction. *Inflamm. Bowel Dis.* 22 (1), 55–67. <https://doi.org/10.1097/MIB.0000000000000592>.
- He-Yang, J., Zhang, W., Liu, J., Xue, P., Zhou, X., 2020. Human breast milk oligosaccharides attenuate necrotizing enterocolitis in rats by suppressing mast cell accumulation, DPPI activity and TLR4 expression in ileum tissue, and regulating mitochondrial damage of Caco-2 cells. *Int. Immunopharm.* 88, 106881. <https://doi.org/10.1016/j.intimp.2020.106881>.
- Heirangkhongjam, M.D., Ngaseppam, I.S., 2018. Traditional medicinal uses and pharmacological properties of *Rhus chinensis* Mill.: a systematic review. *Eur. J. of Integr. Med.* 21, 43–49. <https://doi.org/10.1016/j.eujim.2018.06.011>.
- Hunter, C.J., De Plaen, I.G., 2014. Inflammatory signaling in NEC: role of NFkB and cytokines. *Pathophysiology* 21 (1), 55–65. <https://doi.org/10.1016/j.pathophys.2013.11.010>.
- Isani, M., Illingworth, L., Herman, E., Schmidt, M., Barron, L., Bowling, J., Elizee, M., Bai, I., Gayer, C., Grishin, A., 2018. Soybean-derived recombinant human epidermal growth factor protects against experimental necrotizing enterocolitis. *J. Pediatr. Surg.* 53 (6), 1203–1207. <https://doi.org/10.1016/j.jpedsurg.2018.02.084>.
- Isik, S., Tayman, C., Cakir, U., Koyuncu, I., Turkmenoglu, T.T., Cakir, E., 2019. Sumac (*Rhus coriaria*) for the prevention and treatment of necrotizing enterocolitis. *J. Food Biochem.* 43 (12), e13068 <https://doi.org/10.1111/jfbc.13068>.
- Jilling, T., Lu, J., Jackson, M., Caplan, M.S., 2004. Intestinal epithelial apoptosis initiates gross bowel necrosis in an experimental rat model of neonatal necrotizing enterocolitis. *Pediatr. Res.* 55, 622–629. <https://doi.org/10.1203/01.PDR.0000113463.70435.74>.
- Jilling, T., Simon, D., Lu, J., Meng, F.J., Li, D., Schy, R., Thomson, R.B., Soliman, A., Arditi, M., Caplan, M.S., 2006. The roles of bacteria and TLR4 in rat and murine models of necrotizing enterocolitis. *J. Immunol.* 177 (5), 3273–3282. <https://doi.org/10.4049/jimmunol.177.5.3273>.
- Jing, Y., Peng, F., Shan, Y., Jiang, J., 2018. Berberine reduces the occurrence of neonatal necrotizing enterocolitis by reducing the inflammatory response. *Exp. Ther. Med.* 16 (6), 5280–5285. <https://doi.org/10.3892/etm.2018.6871>.
- Li, J., Wang, T., Liu, P., Yang, F., Wang, X., Zheng, W., Sun, W., 2021. Hesperetin ameliorates hepatic oxidative stress and inflammation via the PI3K/AKT-Nrf2-ARE pathway in oleic acid-induced HepG2 cells and a rat model of high-fat diet-induced NAFLD. *Food Funct.* 12 (9), 3898–3918. <https://doi.org/10.1039/D0FO02736G>.
- Lim, J.C., Golden, J.M., Ford, H.R., 2015. Pathogenesis of neonatal necrotizing enterocolitis. *Pediatr. Surg. Int.* 31, 509–518. <https://doi.org/10.1007/s00383-015-3697-9>.
- Liu, Q., Xu, D., Jiang, S., Huang, J., Zhou, F., Yang, Q., Jiang, S., Yang, L., 2018. Toll-receptor 9 gene in the black tiger shrimp (*Penaeus monodon*) induced the activation of the TLR–NF-κB signaling pathway. *Gene* 639, 27–33. <https://doi.org/10.1016/j.gene.2017.09.060>.
- Liu, Y., Zhu, L., Fatheree, N.Y., Liu, X., Pacheco, S.E., Tatevian, N., Rhoads, J.M., 2009. Changes in intestinal Toll-like receptors and cytokines precede histological injury in a rat model of necrotizing enterocolitis. *Am. J. Physiol. Gastrointest. Liver Physiol.* 297 (3), G442–G450. <https://doi.org/10.1152/ajpgi.00182.2009>.
- Lu, P., Sodhi, C.P., Jia, H., Shaffey, S., Good, M., Branca, M.F., Hackam, D.J., 2014. Animal models of gastrointestinal and liver diseases. Animal models of necrotizing enterocolitis: pathophysiology, translational relevance, and challenges. *Am. J. Physiol. Gastrointest. Liver Physiol.* 306 (11), G917–G928. <https://doi.org/10.1152/ajpgi.00422.2013>.
- Ma, N., Sun, Y., Yi, J., Zhou, L., Cai, S., 2022. Chinese sumac (*Rhus chinensis* Mill.) fruits alleviate indomethacin-induced gastric ulcer in mice by improving oxidative stress, inflammation and apoptosis. *J. Ethnopharmacol.* 284, 114752. <https://doi.org/10.1016/j.jep.2021.114752>.
- Moss, R., Kalish, L., Duggan, C., Johnston, P., Brandt, M., Dunn, J., Ehrenkranz, R., Jaksic, T., Nobuhara, K., Simpson, B., 2008. Clinical parameters do not adequately predict outcome in necrotizing enterocolitis: a multi-institutional study. *J. Perinatol.* 28 (10), 665–674.
- Nakame, K., Kaji, T., Mukai, M., Shinyama, S., Matsufuji, H., 2016. The protective and anti-inflammatory effects of glucagon-like peptide-2 in an experimental rat model of necrotizing enterocolitis. *Peptides* 75, 1–7. <https://doi.org/10.1016/j.peptides.2015.07.025>.
- Neal, M.D., Sodhi, C.P., Jia, H., Dyer, M., Egan, C.E., Yazji, I., Good, M., Afrazi, A., Marino, R., Slagle, D., 2012. Toll-like receptor 4 is expressed on intestinal stem cells and regulates their proliferation and apoptosis via the p53 up-regulated modulator of apoptosis. *J. Biol. Chem.* 287 (44), 37296–37308. <https://doi.org/10.1074/jbc.M112.375881>.
- Neu, J., 2020. Necrotizing enterocolitis: the future. *Neonatology* 117 (2), 240–244. <https://doi.org/10.1159/000506866>.
- Neu, J., Walker, W.A., 2011. Necrotizing enterocolitis. *N. Engl. J. Med.* 364, 255–264. <https://doi.org/10.1056/NEJMra1005408>.
- Niño, D.F., Sodhi, C.P., Hackam, D.J., 2016. Necrotizing enterocolitis: new insights into pathogenesis and mechanisms. *Nat. Rev. Gastroenterol. Hepatol.* 13 (10), 590–600. <https://doi.org/10.1038/nrgastro.2016.119>.
- Nowicki, P.T., 2005. Ischemia and necrotizing enterocolitis: where, when, and how. *Semin. Pediatr. Surg.* 14 (3), 152–158. <https://doi.org/10.1053/j.sempedsurg.2005.05.003>.
- Ozcan, C.U., Hismiogullari, A.A., Sunay, F.B., Ozcan, T., Berksoy, E.A., Aksoz, E., 2018. Protective effects of quercetin on necrotizing enterocolitis in a neonatal rat model. *Am. J. perinat.* 35 (5), 434–440. <https://doi.org/10.1055/s-0037-1608660>.
- Özdemir, Ö.M.A., Ergin, H., Yenisey, C., Türk, N.Ş., Şimşek, N.G., 2010. Protective effects of clarithromycin in rats with hypoxia/reoxygenation-induced intestinal injury. *J. Pediatr. Surg.* 45 (11), 2169–2174. <https://doi.org/10.1016/j.jpedsurg.2010.06.024>.
- Ross, D., Siegel, D., 2017. Functions of NQO1 in cellular protection and CoQ10 metabolism and its potential role as a redox sensitive molecular switch. *Front. Physiol.* 8, 595. <https://doi.org/10.3389/fphys.2017.00595>.
- Shen, H., Lei, Y., He, X., Liu, D., He, Z., 2019. Role of lactadherin in intestinal barrier integrity in experimental neonatal necrotizing enterocolitis. *J. Cell. Biochem.* 120 (12), 19509–19517. <https://doi.org/10.1002/jcb.29255>.
- Sodhi, C., Richardson, W., Gribar, S., Hackam, D.J., 2008. The development of animal models for the study of necrotizing enterocolitis. *Dis. Models Mech.* 1, 94–98. <https://doi.org/10.1242/dmm.000315>.
- Szliszka, E., Krol, W., 2011. The role of dietary polyphenols in tumor necrosis factor-related apoptosis inducing ligand (TRAIL)-induced apoptosis for cancer chemoprevention. *Eur. J. Cancer Prev.* 20 (1), 63–69. <https://doi.org/10.1097/CEJ.0b013e32833283ec48>.
- Tangpu, V., Yadav, A.K., 2004. Antidiarrhoeal activity of *Rhus javanica* ripen fruit extract in albino mice. *Fitoterapia* 75 (1), 39–44. <https://doi.org/10.1016/j.fitote.2003.08.015>.
- Tayman, C., Aydemir, S., Yakut, I., Serkant, U., Ciftci, A., Arslan, E., Koç, O., 2016. TNF-α blockade efficiently reduced severe intestinal damage in necrotizing enterocolitis. *J. Invest. Surg.* 29 (4), 209–217. <https://doi.org/10.3109/08941939.2015.1127449>.
- Tayman, C., Cekmez, F., Kafa, I.M., Canpolat, F.E., Cetinkaya, M., Uysal, S., Tunc, T., Sarici, S.U., 2012. Beneficial effects of *Nigella sativa* oil on intestinal damage in necrotizing enterocolitis. *J. Invest. Surg.* 25 (5), 286–294. <https://doi.org/10.3109/08941939.2011.639849>.
- Van Belkum, M., Alvarez, L.M., Neu, J., 2020. Preterm neonatal immunology at the intestinal interface. *Cell. Mol. Life Sci.* 77 (7), 1209–1227. <https://doi.org/10.1007/s00181-019-03316-w>.
- Wang, K., Tao, G., Sylvester, K.G., 2019. Recent advances in prevention and therapies for clinical or experimental necrotizing enterocolitis. *Dig. Dis. Sci.* 64, 3078–3085. <https://doi.org/10.1007/s10620-019-05618-2>.
- Wu, Z., Ma, Y., Gong, X., Zhang, Y., Zhao, L., Cheng, G., Cai, S., 2019. *Rhus chinensis* Mill. fruits prevent high-fat/ethanol diet-induced alcoholic fatty liver in rats via AMPK/SREBP-1/FAS signaling pathway. *J. Funct. Foods* 61, 103498. <https://doi.org/10.1016/j.jff.2019.103498>.
- Wu, Z., Zhang, Y., Gong, X., Cheng, G., Pu, S., Cai, S., 2020. The preventive effect of phenolic-rich extracts from Chinese sumac fruits against nonalcoholic fatty liver disease in rats induced by a high-fat diet. *Food Funct.* 11, 799–812. <https://doi.org/10.1039/C9FO02262G>.
- Xiao, S., Li, Q., Hu, K., He, Y., Ai, Q., Hu, L., Yu, J., 2018. Vitamin A and retinoic acid exhibit protective effects on necrotizing enterocolitis by regulating intestinal flora and enhancing the intestinal epithelial barrier. *Arch. Med. Res.* 49 (1), 1–9. <https://doi.org/10.1016/j.arcmed.2018.04.003>.
- Yakut, H.I., Koyuncu, E., Cakir, U., Tayman, C., Koyuncu, İ., Turkmenoglu, T.T., Cakir, E., Ozazici, A., Aydogan, S., Zenciroglu, A., 2020. Preventative and therapeutic effects of fennel (*Foeniculum vulgare*) seed extracts against necrotizing enterocolitis. *J. Food Biochem.* 44 (8), e13284 <https://doi.org/10.1111/jfbc.13284>.
- Yang, M., Jiang, Z.-h., Li, C.-g., Zhu, Y.-j., Li, Z., Tang, Y.-z., Ni, C.-l., 2018. Apigenin prevents metabolic syndrome in high-fructose diet-fed mice by Keap1-Nrf2 pathway. *Biomed. Pharmacother.* 105, 1283–1290. <https://doi.org/10.1016/j.biopha.2018.06.108>.
- Yang, Y., Zhang, T., Zhou, G., Jiang, X., Tao, M., Zhang, J., Zeng, X., Wu, Z., Pan, D., Guo, Y., 2020. Prevention of necrotizing enterocolitis through milk polar lipids reducing intestinal epithelial apoptosis. *J. Agric. Food Chem.* 68 (26), 7014–7023. <https://doi.org/10.1021/acs.jafc.0c02629>.
- Yurttutan, S., Ozdemir, R., Canpolat, F.E., Oncel, M.Y., Uysal, B., Unverdi, H.G., Erdevi, O., Dilmen, U., 2013. Protective effects of colchicine in an experimental model of necrotizing enterocolitis in neonatal rats. *J. Surg. Res.* 183 (1), 156–162. <https://doi.org/10.1016/j.jss.2013.01.002>.

- Zani, A., Cordischi, L., Cananzi, M., De Coppi, P., Smith, V., Eaton, S., Pierro, A., 2008. Assessment of a neonatal rat model of necrotizing enterocolitis. *Eur. J. Pediatr. Surg.* 18 (6), 423–426. <https://doi.org/10.1055/s-2008-1038951>.
- Zhang, C., Ma, Y., Zhao, Y., Hong, Y., Cai, S., Pang, M., 2018. Phenolic composition, antioxidant and pancreatic lipase inhibitory activities of Chinese sumac (*Rhus chinensis* Mill.) fruits extracted by different solvents and interaction between myricetin-3-O-rhamnoside and quercetin-3-O-rhamnoside. *Int. J. Food Sci. Technol.* 53 (4), 1045–1053. <https://doi.org/10.1111/ijfs.13680>.
- Zhang, D., Wen, J., Zhou, J., Cai, W., Qian, L., 2019. Milk fat globule membrane ameliorates necrotizing enterocolitis in neonatal rats and suppresses lipopolysaccharide-induced inflammatory response in IEC-6 enterocytes. *J. Parenter. Enteral Nutr.* 43 (7), 863–873. <https://doi.org/10.1002/jpen.1496>.
- Zhang, Y., Wang, O., Ma, N., Yi, J., Mi, H., Cai, S., 2021. The preventive effect and underlying mechanism of *Rhus chinensis* Mill. fruits on dextran sulphate sodium-induced ulcerative colitis in mice. *Food Funct.* 12, 9965–9978. <https://doi.org/10.1039/D1FO01558C>.
- Zhou, B.-h., Tan, P.-p., Jia, L.-s., Zhao, W.-p., Wang, J.-c., Wang, H.-w., 2018. PI3K/AKT signaling pathway involvement in fluoride-induced apoptosis in C2C12 cells. *Chemosphere* 199, 297–302. <https://doi.org/10.1016/j.chemosphere.2018.02.057>.